

aaronbellkushaiedit

Travel Radio Australia and we're at the Disaster and Emergency Management Conference here on the Gold Coast and I've got here Aaron Bell from Kushai and Kushai are a fascinating company. They make hardware, trailers, they do drones, they do AI and they work with major organisations in Australia such as Surf Life Saving Australia. Now Aaron, welcome to the program.

Hello and thanks for letting me be here. Yes, now Aaron tell us about Kushai, what basically does it do and how does it work? In basic terms we are a technology integrator but we work at the intersection of drones and AI machine learning and any kind of what we call visual analytics. So if it's a camera we can analyse the footage that it's either recording or displaying and then teach an AI model to find things that we need to find.

That could be a surfer in distress or looking for sharks, it could be looking for feral animals, it could be looking for plastics on the beaches up in Torres Strait and one of the first projects we ever started on was finding rock art in far north Queensland. So we're five and a half hours northwest of Cairns in three million acres of untouched wilderness in an area of three million acres that's under assessment for a UNESCO World Heritage Site and there are estimated to be 50,000 rock art sites out there and they didn't have any of them documented. So we figured out a way that we could use autonomous systems to fly around, some AI to detect the rock art and then put it on a record so that we know where it is and how to protect these ancient sites.

So that's where we started. Yeah that's incredible and these rock art places are visible from the air and it's so clever isn't it because you wonder how the Aboriginal communities tens of thousands of years ago were so smart to actually do all this. Yeah absolutely and that's part of one of the things where we started is that there has been a gap so with the stolen generation a lot of that knowledge has been lost and so we went in and worked with our brothers and sisters up there and started to document it and that's where it started and now it's migrated to surf lifesaving and RFS and a few of those kind of applications.

Yeah I'm looking at your trailer here and you have a beautiful four-wheel drive with all the amazing technology built into the four-wheel drive and the trailer which you can just park up anywhere on the beach and it's got the drone facilities there that can go shark hunting and all that sort of thing. Indeed so what we had to do was integrate quite a few different bits of technology. So firstly the drone well that can fly but we use a thing called a drone docks or a drone in a box so it's like a little clamshell or a pod or a hive where the drone sits in and charges and uploads its data but that hive then needs power so we had to find batteries and solar and then of course we needed to communicate so we had to set some internet up so we're using low orbit satellites so things like Starlink so it can now communicate with the internet and then of course as I said we work with both the hardware the drones and the trailer and the AI so then we use edge compute so the trailer comes with computation power that we can then process the data that it's collecting

and then beam it all back to to the headquarters.

Yes AI has really taken over here hasn't it it's really in these situations for emergency management it's absolutely crucial and cuts down your work incredibly. It does so this is what I love about it is that you know sometimes people can be a bit sceptical about it but in this case it's a really really wholesome community-led applications of this kind of technology so finding sharks etc is the first use case that we're talking about now what it enables us to do is that a pilot that's flying the drone can be looking at the screen be they remotely so let's say they're in Brisbane and the drone was on the Gold Coast they can be flying it remotely watching the screen they could be on the beach looking at the screen on the drone that they're flying what the AI does will help them detect more effectively whatever it is that they're looking for a shark a rock fisherman that's been washed into the water a rip or somebody that might be in distress. What about rural areas I suppose you go into really remote rural areas and and what what sort of applications are out there? That was one of the biggest things that we sort of really had to work on particularly when we started with that rock art piece and that's where the trailer came from is we realised that when you're in the middle of the cape and you're two three four five hours away from the nearest large town you don't necessarily have 5G there's no mobile phones there's no 240 volt power so we had to figure out a way that we could operate in those remote areas so that's where the solar and the battery and the starlinks all came in and then the applications so drones are really good at covering large areas and getting quite detailed information so we could do things from checking culverts on a road or counting wildlife finding a herd we can they're really good at fence inspections or again in those remote areas up in the Torres Strait they use the blue water highway there's no roads because it's all islands so when kids are going to school in a small dinghy and that dinghy flips how do you find them quickly you can't get a helicopter that's two hours away to do that so this kind of technology can sit on a remote island it can monitor areas where of interest where things hazards might occur or where we're worried about something that we need to look for so if someone's uh for instance if someone's out on a lonely highway somewhere with no facilities around you can send out a drone you can locate them you can pinpoint where they are that's very useful without getting a massive helicopter out there at massive expense that's exactly right so i mean these drones they cost about ten thousand dollars each so they're not it's not a cheap toy but when you it costs about ten thousand dollars an hour to fly a helicopter the drone makes a lot more sense on top of that the barrier to entry so you need to spend quite a lot of time and money to learn how to fly a helicopter i could train my grandma to fly a drone in about 10 minutes so the barrier to entry is a lot lower and so what you do is that you know they can't cover the same distance as an aircraft but what you can do is they fit into the system better so you have many drones to a few helicopters and a few helicopters to one really big expensive airplane and those many drones can then help authorities such as the rural fire service for example direct their most precious assets to the area of greatest need the name kushai which is k-u-s-h-a-i dot a-i is the website where did the name come from i mentioned we do quite a lot of work in far north queensland and with a lot of indigenous

communities and one of our favorite indigenous brothers was we're driving up the pdr on our way up to satia trying to come up with a company name and he said what about the kushai man i said what's the kushai man and the kushai man is a highly respected elder a keeper of traditional law and that speaks truth to all of the clans and and make sure that the the story's understood and it's consistent and that's why there's so much consistency with the aboriginal song lines right across the country and we like the idea that we can travel remotely we're using drones and technology to be able to provide good information and sound data like sound law and consistent law and consistent data to make good decisions and so it was a really good fit it appealed to our concern for our first nations it appealed to our technology it appealed to the way we operate in remote areas so it's a great fascinating story so if people want to go there please go there kushai.ai k-u-s-h-a-i.ai fascinating stories and these drones when we say drones these are serious pieces of kit they're not like your little 249 gram drones that the kids play with no they're not they're they're a little bigger they're not they're not huge you know they're only sub two kilos most of the time we have a few that are a touch bigger but at that that point we have thermal imaging they've got laser range finding they can run ai at the edge so the drones are actually capable of running an ai model without even being connected to the internet they have very very sophisticated ways video share we can share location data and i can drop a pin with my drone and then share that with police or the firemen and and let them know that this is the exact address of where where the incidents occurring and speed up responses thank you so much aaron bell for joining us from kushai it's been fascinating and thanks for being part of the program it's my pleasure thank you